Which mode are you in? Lesson notes

Speaker notes for the slides, one section per slide, with timings. About 50 minutes in all.
Print this for yourself; the slides are in lesson-slides.pdf.

Slide 1 · Which mode are you in?

1 minute. Before the lesson: print one set of stance cards per table, and one self-check per student. Prepare the slide 12 activity: print activity-handout.pdf (one half-page per pair), or make your own example (see the slide 12 notes). Students do not need a device or an account for any part of this lesson.

Timings add up to 42 minutes, plus about 8 for the self-check at the end: 50 in all.

Say: "Today is not about whether you're allowed to use AI. You already use it, or you will. It's about something nobody has taught you yet: what kind of help you're asking for, and who checks it."

Slide 2 · The calculator

2 minutes. Ask first: "Hands up if you think calculators made people worse at maths." Take two answers.

Then say: "When calculators came into classrooms, people had exactly the argument we're having about AI. In 1980 the US maths teachers' council recommended using them at every level, and making problem solving the focus. The machine took the arithmetic, and the human curriculum moved up to the harder thinking."

The quotation is from the National Council of Teachers of Mathematics, *An Agenda for Action* (1980): programs should "take full advantage of the power of calculators and computers at all grade levels".

Slide 3 · Foundations first, then the tool

2 minutes. Say: "Researchers looked at dozens of studies afterwards. Calculators did not hold students back when they were part of how maths was taught and tested. The one harm signal was early: sustained use in Grade 4 'appears to hinder the development of basic skills in average students', in the researchers' words. Foundations first, then the tool."

Point at the bold line. "That's the question for today. Not 'should we use AI?' but: for which problems, when, and who checks?"

Sources: Ellington (2003), a meta-analysis of 54 studies; Hembree and Dessart (1986), a meta-analysis of 79 research reports. Both in the Journal for Research in Mathematics Education.

Slide 4 · Where the comparison breaks

2 minutes. Say: "The calculator is a flattering comparison. Type 7 times 8 and you get 56, every time. An AI is different. It can give you a wrong answer that sounds exactly like a right one: same confidence, same tidy sentences. And it doesn't stay below the problem the way a calculator does. It can do the framing, the reasoning and the writing, which is the part you're here to learn."

Ask: "Has anyone had an AI tell you something confidently that turned out to be wrong?" Take one example. If nobody has, say: "You may not have noticed. That's the point."

Slide 5 · Same AI, different results

4 minutes. This is the most important evidence in the lesson. Take it slowly.

Say: "Researchers gave about a thousand high-school maths students one of two versions of the same AI, or no AI. One version answered questions, like a normal chatbot. The other was set up to give hints and ask questions, but not the answer."

Reveal the practice column. "With AI, practice scores went up for both. The answer version by about 48%, the hint version by about 127%."

Then the exam column. "Then they sat the exam without AI. The students who had practised with the answer version did about 17% worse than students who never had AI at all. For the hint group, the researchers found no significant difference from students without AI."

Ask: "Why would a version that raised practice scores make the exam worse?" Let them get there: the answer version did the thinking for them. Then read the bold line: unrestricted answers hurt the exam; a tutor with guardrails largely removed the harm. Same AI; the kind of help made the difference. Do not say the hint version made students better at the exam: it did not hurt them, which is a different claim.

Source: Bastani, Bastani, Sungu, Ge, Kabakcı and Mariman, "Generative AI without guardrails can harm learning: Evidence from high school mathematics", *PNAS* 122(26), July 2025.

Slide 6 · Before you type: what are you expecting?

2 minutes. Hand out the stance cards now, one set per table.

Say: "Every time you open an AI, you have an expectation before you type: what do you want back? That expectation is your stance. Most people never name it. The same words can mean three different things, and get three very different results for your learning."

Ask: "Which of the three on the slide do you do most?" Quick show of hands, no judgement. Then: "There are five stances on your table. Let's go through them."

Slide 7 · The Oracle

3 minutes. Say: "The Oracle is the answer machine. It's where almost everybody starts. The trouble is that a fluent wrong answer looks exactly like a fluent right one, and if you asked because you didn't know, you can't tell the difference."

Read the rule aloud. "A claim is something you check. A verdict is something you accept."

Made-up example to give: a student revising for history asks for the three causes of a war and the date it ended. The answer reads well and the date is a year out. The student who asked "which parts are you least sure of?" and checked the textbook catches it.

Slide 8 · The Tutor

3 minutes. Say: "The Tutor is the one the hint version in that study was built to be. It's slower than the Oracle. That's the point: the effort is where the learning happens."

"The way it fails: if you don't say what you're aiming for, it will teach you whatever it guesses you need. So tell it first: what do you want to be able to do afterwards? A test on Friday, without a calculator? An essay you can defend in your own words?"

"If you remember one sentence from today, make it the last two on this slide: Don't tell me the answer yet. What questions should I be asking?"

Slide 9 · The Mirror

2 minutes. Say: "AI models may simply agree with whoever is typing. Ask 'is my essay good?' and you may hear yes whether it is or not. That isn't feedback; it's flattery."

"The Mirror works when you ask for the shape of your own thinking: your claims, in order, with the evidence you gave. Then you can see for yourself that claim three has no evidence at all."

Slide 10 · The Adversary

2 minutes. Say: "This one takes nerve: you ask the AI to find the holes in your work. If you just say 'criticise this', you'll get comments on your spelling, or on a point you never meant to make."

"So tell it what you're defending, and what would make you change your mind. Then it can go for the weak link." Good use: before a debate, before handing in an argument, before an oral.

Slide 11 · The Librarian

2 minutes. Say: "The Librarian is the AI as a research assistant: what have people said about this? It's useful, and it has a known failure: sources that don't exist. Real authors with books they never wrote. Real books with quotes that aren't in them."

"The rule is simple and nobody does it: before you use a source, look it up. Does it exist? Does it say what the AI said it says?"

Slide 12 · Fluently wrong

7 minutes. Prepare before the lesson. You have two options.

Ready-made: print activity-handout.pdf, one half-page per pair. It holds a short answer about why the United States joined the First World War and how the war ended. It is a made-up illustration written for this pack, not real AI output, in the confident style an answering AI often has. It contains three errors:

- it says the United States entered the war in April 1916; it was April 1917 (and the Zimmermann Telegram it mentions is from January 1917, so the answer's own timeline does not fit);
- it says the armistice was signed in the Hall of Mirrors at Versailles at 11 a.m.; it was signed in a railway carriage near Compiègne in the early morning and took effect at 11 a.m. (the Hall of Mirrors is where the Treaty of Versailles was signed, in June 1919);

- its source, "M. Hartwell, *Empires at the Brink*", was invented for this illustration; students should not find it.

The guide's appendix has the full text and this key. The topic is only an example: swap in a short answer from your own subject that you have checked is wrong; the three checks stay the same.

Your own: with an AI tool your school has approved, ask a question in your subject whose answer students can check from their textbook or notes: a set of dates and causes, a worked solution, or "give me three sources with page numbers on [a topic you know well]". Ask in the Oracle stance, just the question. Use an answer that contains at least one error you have checked yourself; it may take a few tries. If your school has no approved tool, use the ready-made handout.

In class: pairs, one handout each. Five minutes on the three steps, two minutes of feedback. Ask: "What did you check it against?" and "Would you have noticed if you hadn't been told to look?" For the third step: the question was asked in the Oracle stance; what should have come first is "Before you answer, tell me which parts you are least sure about", and for the source, the Librarian's rule.

If you used your own and the answer turns out to be entirely right, use it anyway and ask: "How do you know it's right?" The answer is the same: you checked.

Students do not use an AI themselves in this activity, so no student needs an account or a device.

Slide 13 · Who checks?

3 minutes. Say: "Asking the same chat 'are you sure?' is not a check. The AI that gave you the wrong date may simply agree with itself, or change its answer because you sounded doubtful. Either way, you haven't checked anything. A check is something outside the conversation: your textbook, the original document, your own working, or doing a bit of it yourself and comparing."

"And for your assignments, I'm a checker too. Some assignments will have a part you do without AI, in class or in conversation with me. That part is how I check that you understand your own work."

If asked about detectors, speak for this pack, not for the school: "This pack uses no detector; the unaided part is the check." Background, if useful: a 2023 study found detectors flagged essays by non-native English writers as AI-written far more often than essays by native speakers (Liang et al., *Patterns*, July 2023). Your school's or exam board's own rules still apply.

Slide 14 · The mode log

2 minutes. Hold up a mode log sheet; there is also one at the end of the self-check. Each assignment that asks for a log gets its own sheet.

Say: "When an assignment says 'mode log', you fill one line every time you use AI on it. What you asked, in a few words. Which mode you were in. What you checked yourself. What you changed."

"The log isn't a trap. An honest 'Oracle, checked nothing' is more useful to me than a tidy log that didn't happen, because it tells us both what to work on. In a tutor-only assignment: if you used the Oracle, record it; it counts as a deviation, and I'd rather know. I'll sometimes pick one line and ask you to show me."

Slide 15 · Three assignment modes

3 minutes. Hold up an assignment sheet. Say: "Every assignment this term will come with this sheet, and it says which mode."

"A: no AI, the way you've always worked. That's the foundation, like the non-calculator paper."

"B: Tutor only. The AI can help you understand, question you and give hints. You may also use it as a Mirror, an Adversary or a Librarian, with their rules. Not as an Oracle: you don't ask it for answers. It doesn't write anything you hand in. You keep a mode log."

"C: full use. You can use AI the way people do at work, within the school's rules. You hand in a mode log, a few lines on what the AI did and where you changed it, and you do a short part without AI, so we both know you understand your own work."

Ask: "Which mode do you think will be hardest? Why?"

Slide 16 · The one rule to take away

2 minutes, then the self-check. Say: "If you forget everything else: before you type, say what kind of help you want. Then do that stance's one check. Oracle: ask how sure it is. Tutor: say what you want to be able to do. Mirror: ask for the structure, not agreement. Adversary: say what you're defending. Librarian: look the source up."

"You may use the calculator. You still sign the exam."

Hand out the self-check: about 8 minutes for both pages. If time is short, set it as homework. It is theirs; you do not collect it. The log at its end is for their first logged assignment; hand out a mode log sheet with each later one.